Design and Implementation of a First-Order Sugeno Fuzzy Inference System for Emergency Braking of an Autonomous Vehicle

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Course: Fundamentals of Computational Intelligence

Complete Report

Description:

Suppose an autonomous vehicle is moving and an obstacle suddenly appears in its path. Our system must act in such a way that, using the input data from the sensors, it decides how hard to brake and by how much to turn the steering wheel so that the vehicle does not collide with the obstacle. Because of its mathematical nature and light computation, the Sugeno model makes real-time operation of the vehicle possible in situations like this. Here, the performance of the system has been simulated on 4 critical scenarios.

In the control of autonomous vehicles, traditional algorithms usually have to deal with uncertainty in the environment. Fuzzy logic is an ideal option for modeling such uncertainty. Mamdani fuzzy systems require heavy computation in the defuzzification stage (center-of-gravity calculation), so they are not well suited to embedded systems with limited processing power, and because of their slowness they are not practical for real-time problems. In this project, a first-order Takagi-Sugeno fuzzy system is used. In this system, the output part of each rule, instead of a fuzzy membership function set, is a linear function of the input variables. This property removes the heavy center-of-gravity stage and replaces it with a simple weighted average.

Inputs and Outputs:

Inputs:

1 – Front sensor, range 0 to 10 m (Collision {trapezoidal}, Near {triangular}, Far {trapezoidal})

2 – Right sensor, range 0 to 5 m (Closed {trapezoidal}, Open {trapezoidal})

3 – Left sensor, range 0 to 5 m (Closed {trapezoidal}, Open {trapezoidal})

4 – Current speed, range 0 to 2 m/s (Fast {trapezoidal}, Slow {trapezoidal})

Outputs:

For each rule, two outputs are considered, the steering angle and the braking intensity, each of which is a function of the input variables of the following form:

$$Z = a \cdot Front + b \cdot Right + c \cdot Left + d \cdot Speed + e$$

(in the code the inputs are named Jelo, Rast, Chap and Soraat, respectively)

In this system, 9 separate linear equations have been designed for the steering and 9 equations for the brake. This design makes the control of the vehicle speed-dependent. For example, in collision conditions the speed coefficient (d) is chosen as a large positive or negative value, so that at higher speeds the car steers faster and brakes harder.

1 – Steering angle: range between −45 and +45 degrees

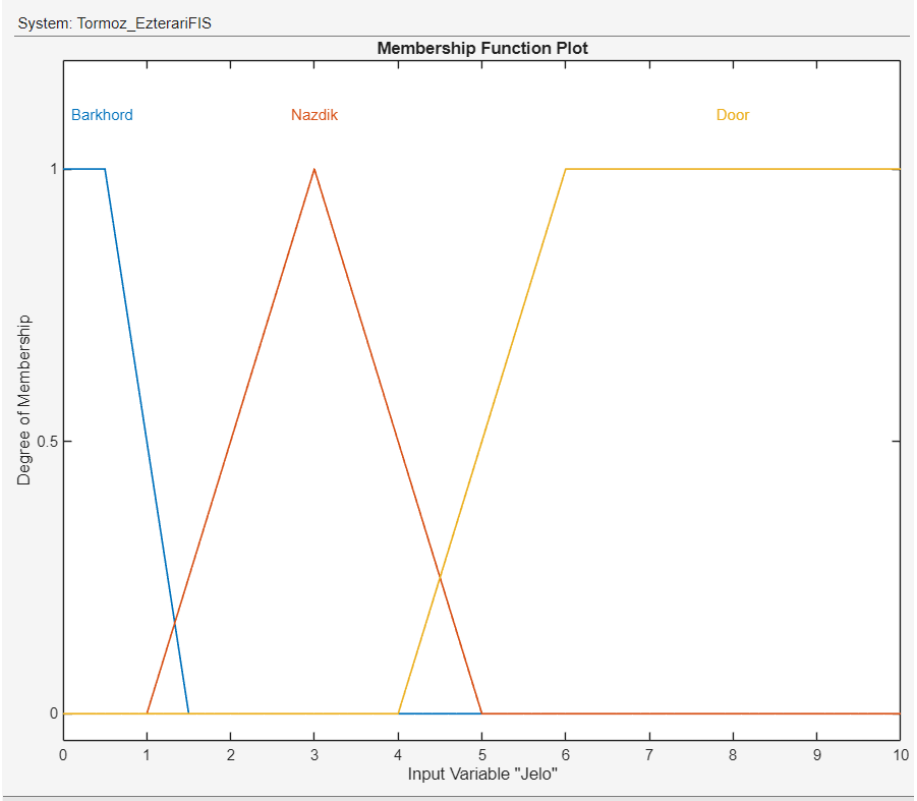
```
CharkheshFarmaanParams = {
    [0  0  0  3  35];    % R1: Tasadof hatmi + Rast baz -> Pish farz 35 daraje be rast + (3 * Soraat)
    [0  0  0 -3 -35];    % R2: Tasadof hatmi + Chap baz -> Pish farz 35 daraje be chap - (3 * Soraat)
    [0  0  0  2  20];    % R3: Tasadof hatmi + Har 2 baz -> Mirim rast ba shibe kamtar
    [0  0  0  0   0];    % R4: Tasadof hatmi + Har 2 baste -> Farman 0 (Faghat tormoz)
    [-3 0  0  5  15];    % R5: Mane nazdik + Rast baz -> Formule dynamic bar asase fasele
    [3  0  0 -5 -15];    % R6: Mane nazdik + Chap baz -> Formule dynamic bar asase fasele
    [0  0  0  0   0];    % R7: Mane nazdik + Har 2 baste -> Farman 0
    [-2 0  0  3  10];    % R8: Mane nazdik + Har 2 baz -> Charkheshe narm be rast
    [0  0  0  0   0];    % R9: Masir door (Amn) -> Farman sefr (Harekate mostaghim)
};
```

2 – Brake: range between 0 and 100 percent

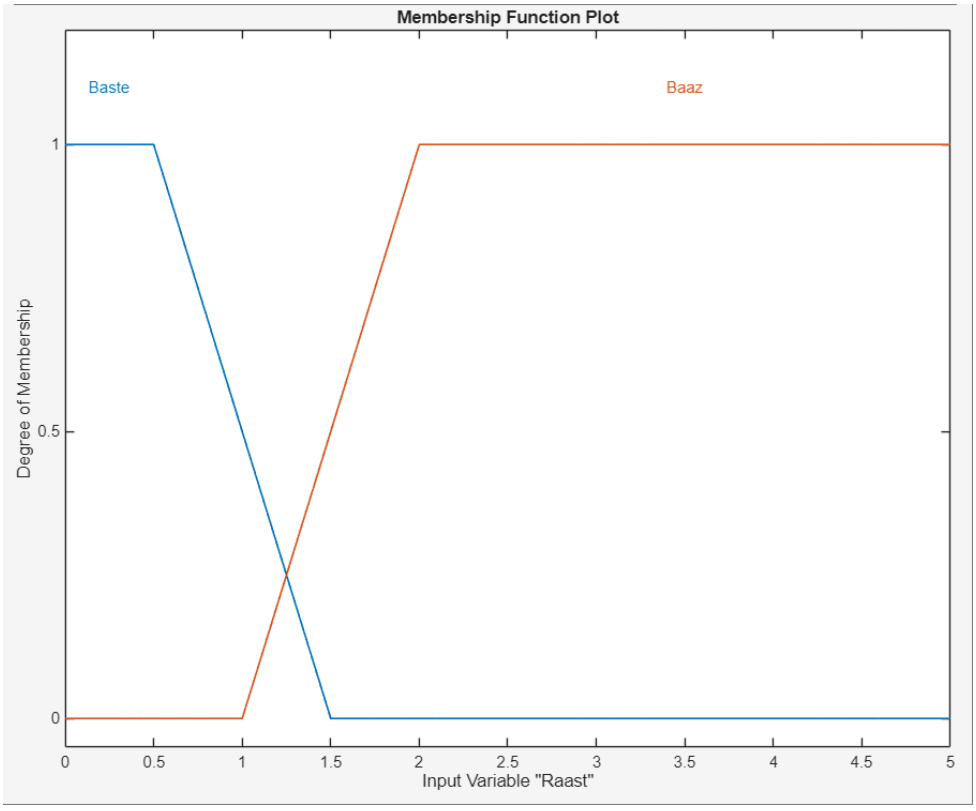
```
TormozParams = {
    [0  0  0 10  80];    % R1: Tormoz shadid 80% + (10 * Soraat)
    [0  0  0 10  80];    % R2: Tormoz shadid
    [0  0  0  8  75];    % R3: Tormoz shadid
    [0  0  0  0 100];    % R4: Tormoze 100% va kamel (Taleh)
    [-2 0  0  8  30];    % R5: Tormoze narmtar chon masir baaz ast
    [-2 0  0  8  30];    % R6: Tormoze narmtar
    [-3 0  0 10  60];    % R7: Tormoze nimeshadid
    [-1 0  0  5  25];    % R8: Tormoze khafif
    [0  0  0  2   5];    % R9: Ye nishtormoz age sorat kheyli bala bood
};
```

Membership function plots of the inputs:

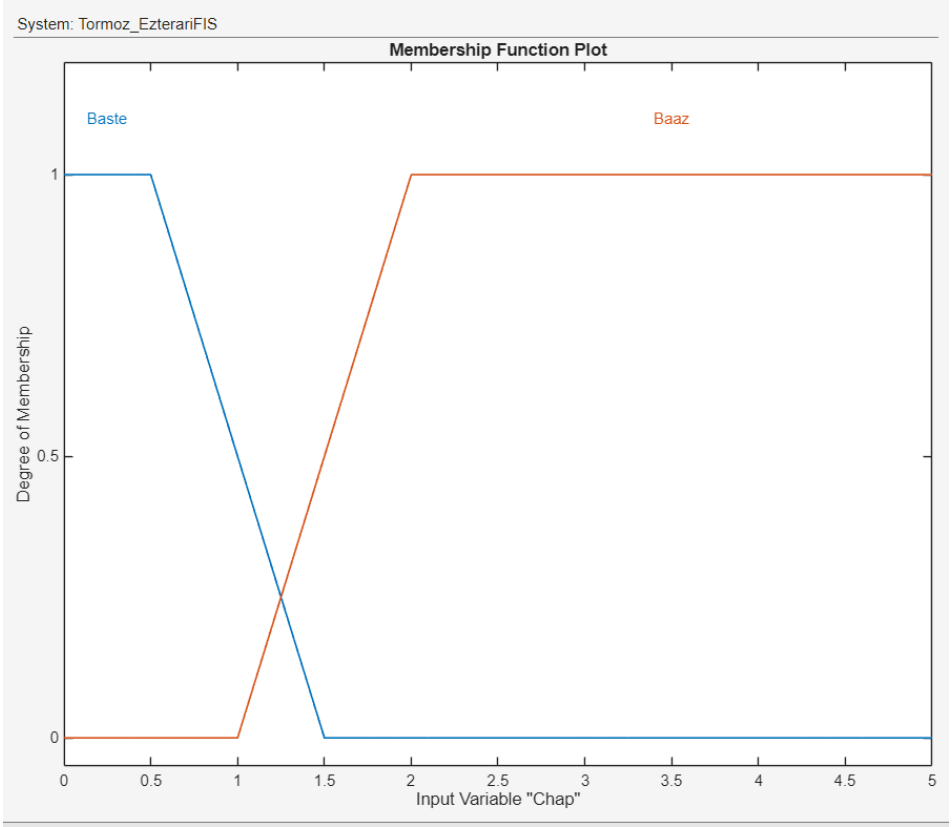
Front sensor



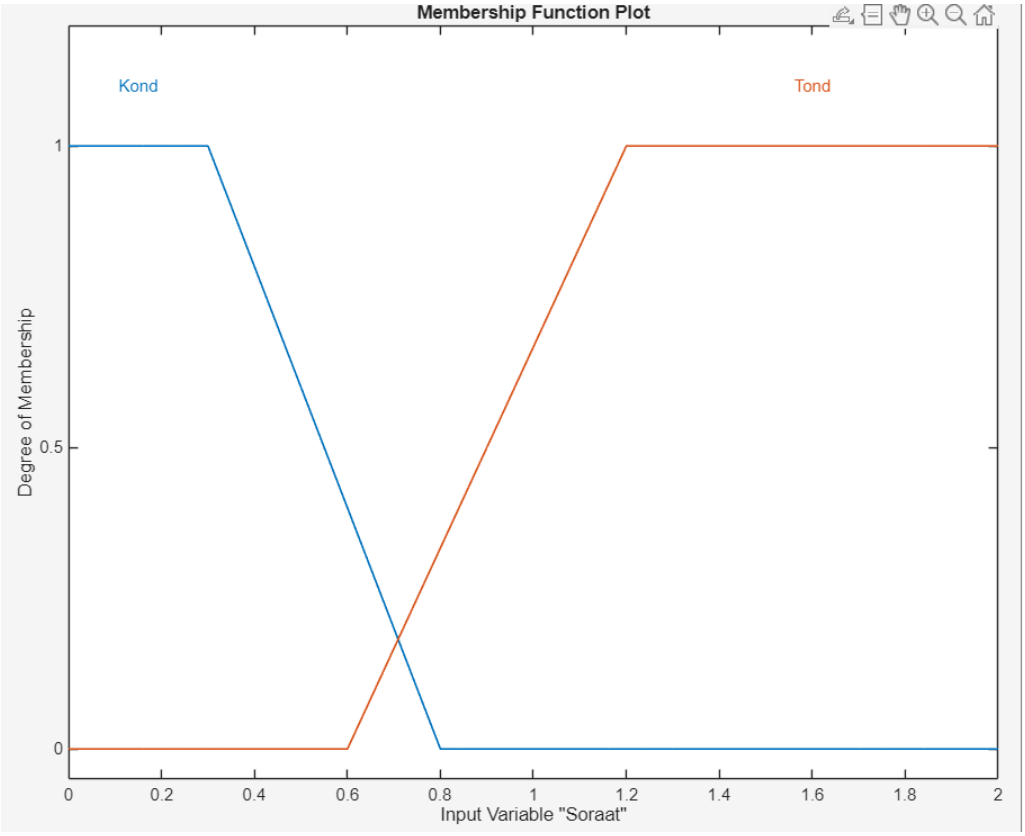
Right sensor



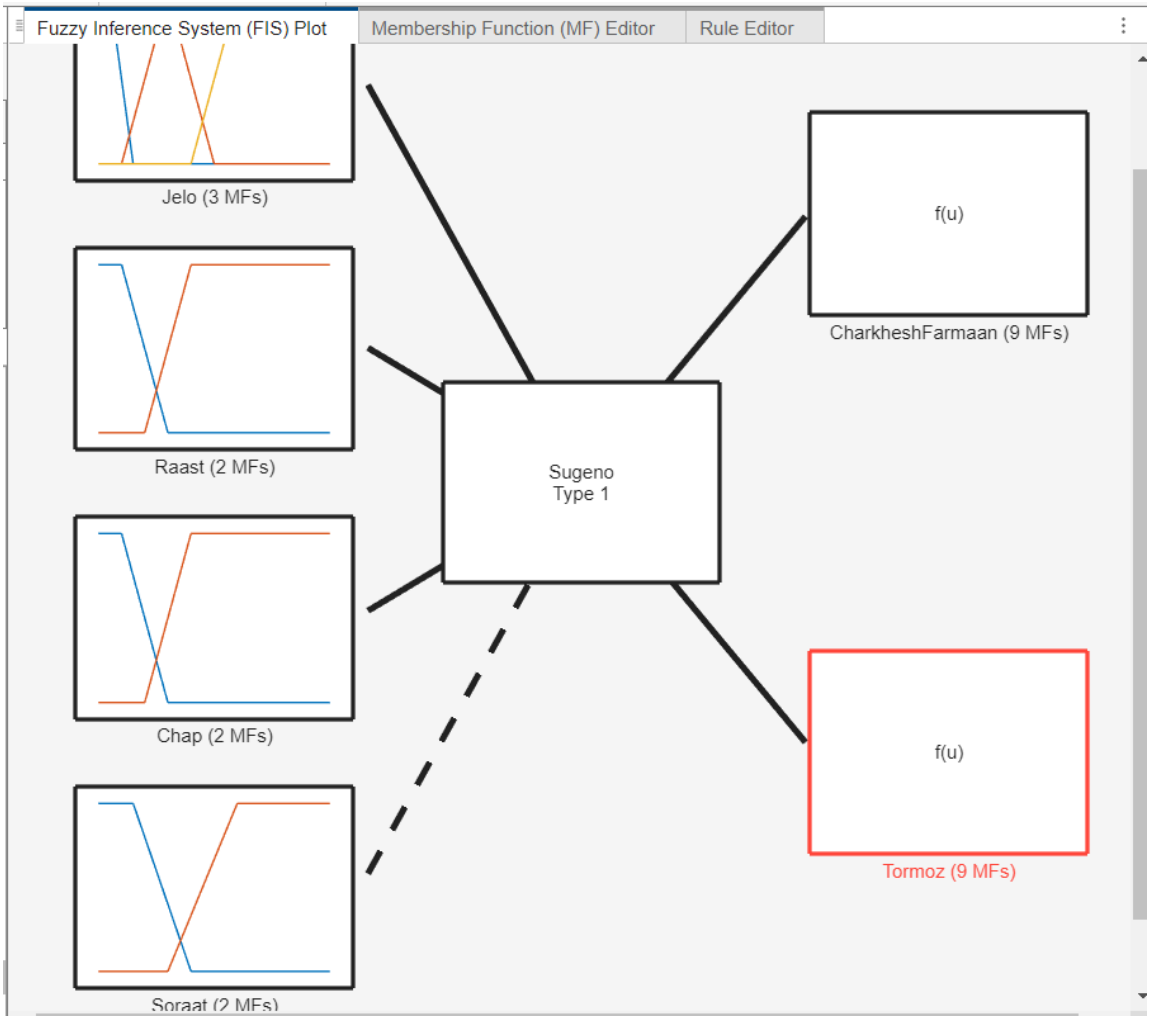
Left sensor



Speed sensor



Overview:



Fuzzy Rules:

Rules 1 to 4 (critical situation): The focus is on severe braking and sudden change of direction, because the distance to the obstacle is very small.

Rules 5 to 8 (controllable situation): The focus is on soft maneuvers, because the car still has time and the obstacle is still at a near distance.

Rule 9 (safe situation): Written to save processing; when the road is open, the car simply accelerates, regardless of the sensors.

Rule	Condition	Action
R1	Front blocked, right open, left blocked	**Escape right:** since the only way out is to the right, the steering turns sharply to the right and heavy braking is applied.
R2	Front blocked, right blocked, left open	**Escape left:** exactly the opposite of the first rule; the steering turns sharply to the left so that the car does not hit the obstacle on the right or the one ahead.
R3	Front blocked, both sides open	**Optimal path selection:** the way ahead is blocked but left and right are both open, so the car chooses to turn right by default.
R4	Front blocked, both sides blocked	**Trap (dead-end) situation:** there is no escape route; the steering is locked at zero and 100% braking is activated, so that only the severity of the impact is reduced.
R5	Front near, right open, left blocked	**Soft lane change to the right:** the obstacle is not yet very close, so the car leans toward the right with mild braking and a soft steering angle.
R6	Front near, right blocked, left open	**Soft lane change to the left:** mild braking together with soft steering to the left in order to pass beside the obstacle.
R7	Front near, both sides blocked	**Precautionary speed reduction:** the front is near and the sides are blocked; the car applies the brake fully to reduce its speed.
R8	Front near, both sides open	**Free maneuver:** the obstacle is in the middle of the road but both sides are open. The car very softly changes its path to the right in order to get past the obstacle.
R9	Front completely far and open	**Highway (free speed):** when the front is open, the left, the right and the speed do not matter at all; the steering is zero and the braking is at its minimum.

The fifth and sixth columns of the rule list below mean: when this rule fires, go to linear equation number so-and-so. In effect, these rules decide which linear equation is applied in each of the two outputs, the brake and the steering angle.

```
% Format: [Jelo Rast Chap Soraat | farman Tormoz | Vazn | AND(1)]
% Adad 0 dar voroodi = "Mohem Nist" (Niyazi be barresie in sensor nist)
ruleList = [
    1  2  1  0   1  1   1  1;   % R1: Jelo(Barkhord), Rast(Baaz), Chap(Baste)
    1  1  2  0   2  2   1  1;   % R2: Jelo(Barkhord), Rast(Baste), Chap(Baaz)
    1  2  2  0   3  3   1  1;   % R3: Jelo(Barkhord), Rast(Baaz), Chap(Baaz)
    1  1  1  0   4  4   1  1;   % R4: Jelo(Barkhord), Rast(Baste), Chap(Baste)
    2  2  1  0   5  5   1  1;   % R5: Jelo(Nazdik), Rast(Baaz), Chap(Baste)
    2  1  2  0   6  6   1  1;   % R6: Jelo(Nazdik), Rast(Baste), Chap(Baaz)
    2  1  1  0   7  7   1  1;   % R7: Jelo(Nazdik), Har do baste
    2  2  2  0   8  8   1  1;   % R8: Jelo(Nazdik), Har do baaz
    3  0  0  0   9  9   1  1;   % R9: Jelo(Door) -> Baghiye sensorha 0 (mohem nistan)
];
```

Defuzzification Process:

In Sugeno systems, the defuzzification stage follows the weighted-average formula. If the firing strength of the i-th rule is W_i (the result of the AND operator, i.e. the minimum of the membership degrees) and the linear output of that rule is Z_i, the final output is obtained from the following relation:

$$\frac{\sum_{i=1}^{n} w_i \cdot Z_i}{\sum_{i=1}^{n} w_i}$$

In what follows, the way this system works is explained with several pre-designed scenarios (inputs are given in the order [Front, Right, Left, Speed]):

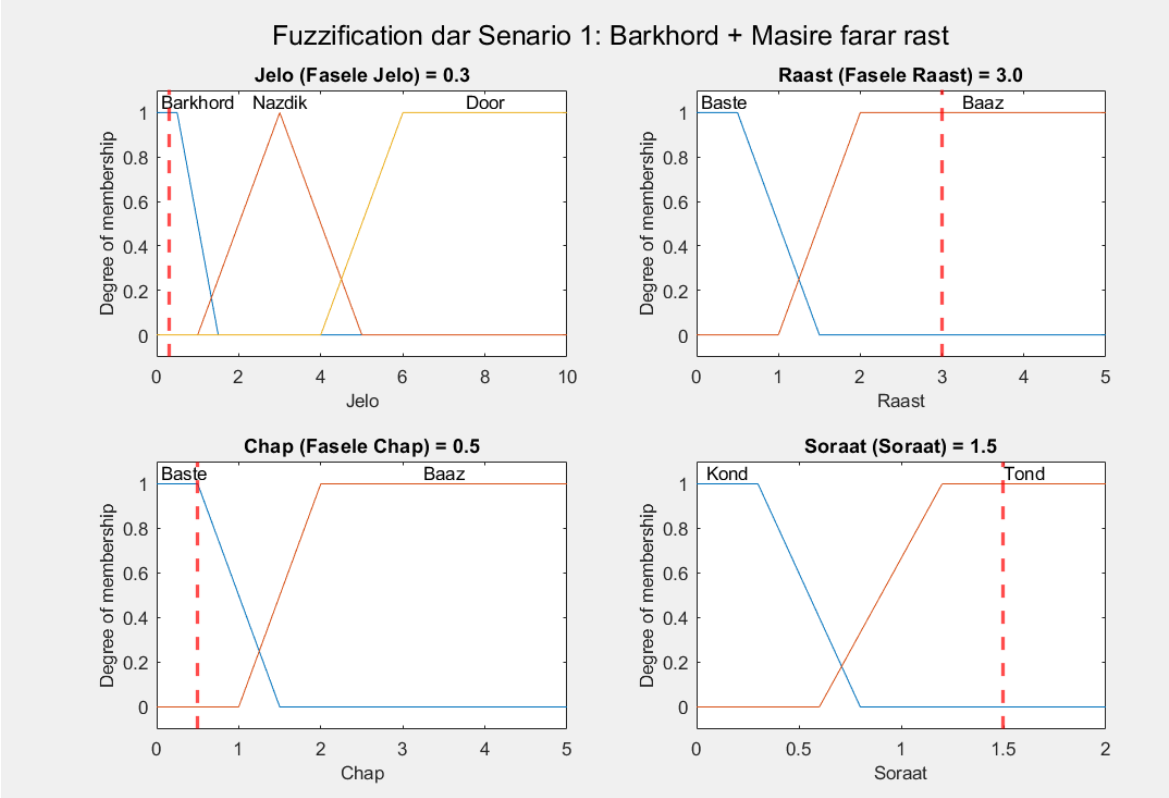
1) **Scenario 1 (certain collision, escape route on the right):** the inputs are [0.3, 3.0, 0.5, 1.5]. In this case, severe braking is activated at a high rate and the steering turns sharply toward the right so that the robot escapes a direct collision.

2) **Scenario 2 (complete encirclement):** the inputs are [0.3, 0.5, 0.5, 0.5]. Because all routes are closed, the system issues a 100% brake command and locks the steering angle at zero degrees (maximum stability).

3) **Scenario 3 (obstacle at medium distance):** the inputs are [3.0, 3.0, 0.5, 1.0]. A gentle steering angle toward the right is applied and the brake pressure increases in a controlled, soft manner.

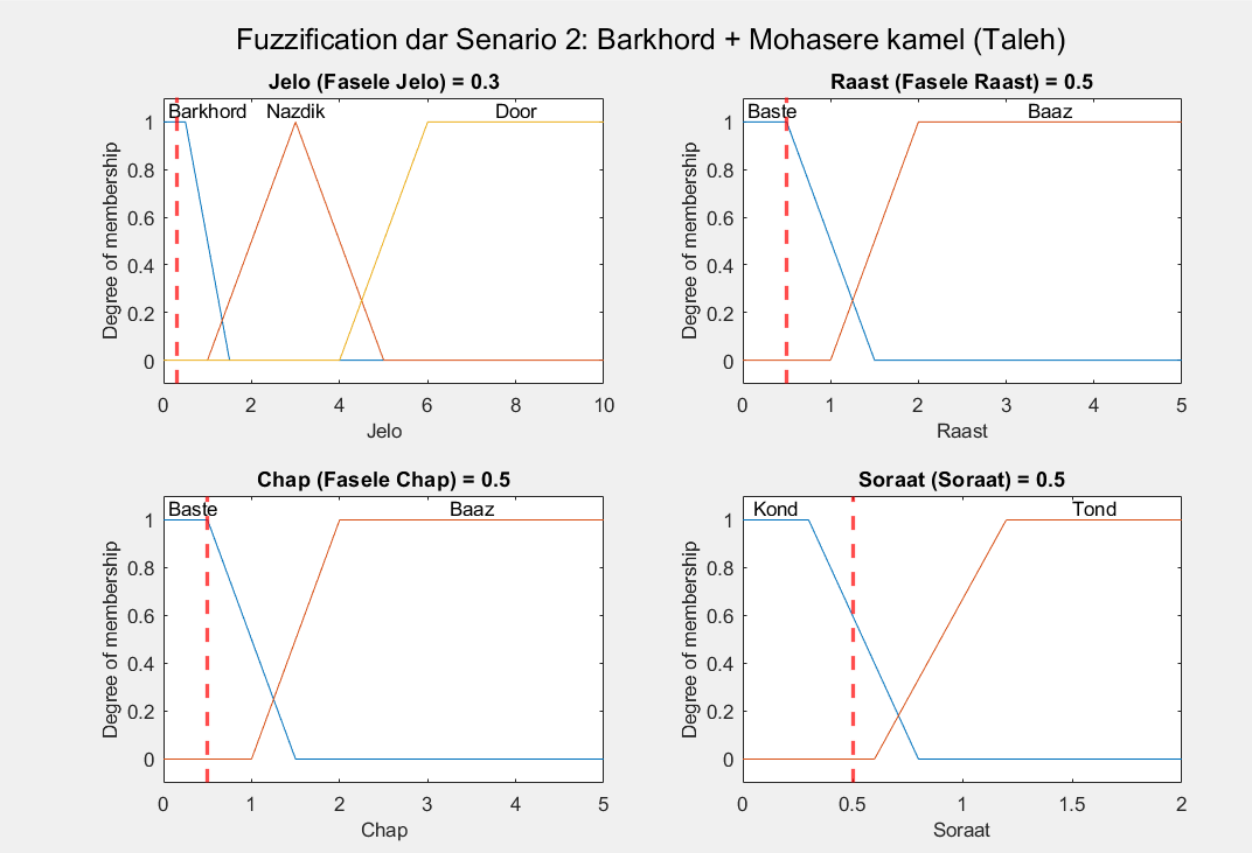
4) **Scenario 4 (highway and completely open path):** the inputs are [8.0, 1.0, 1.0, 1.8]. Since rule R9 is activated, the robot continues along its straight path with a zero steering angle and the brake at its lowest level.

Scenario 1:



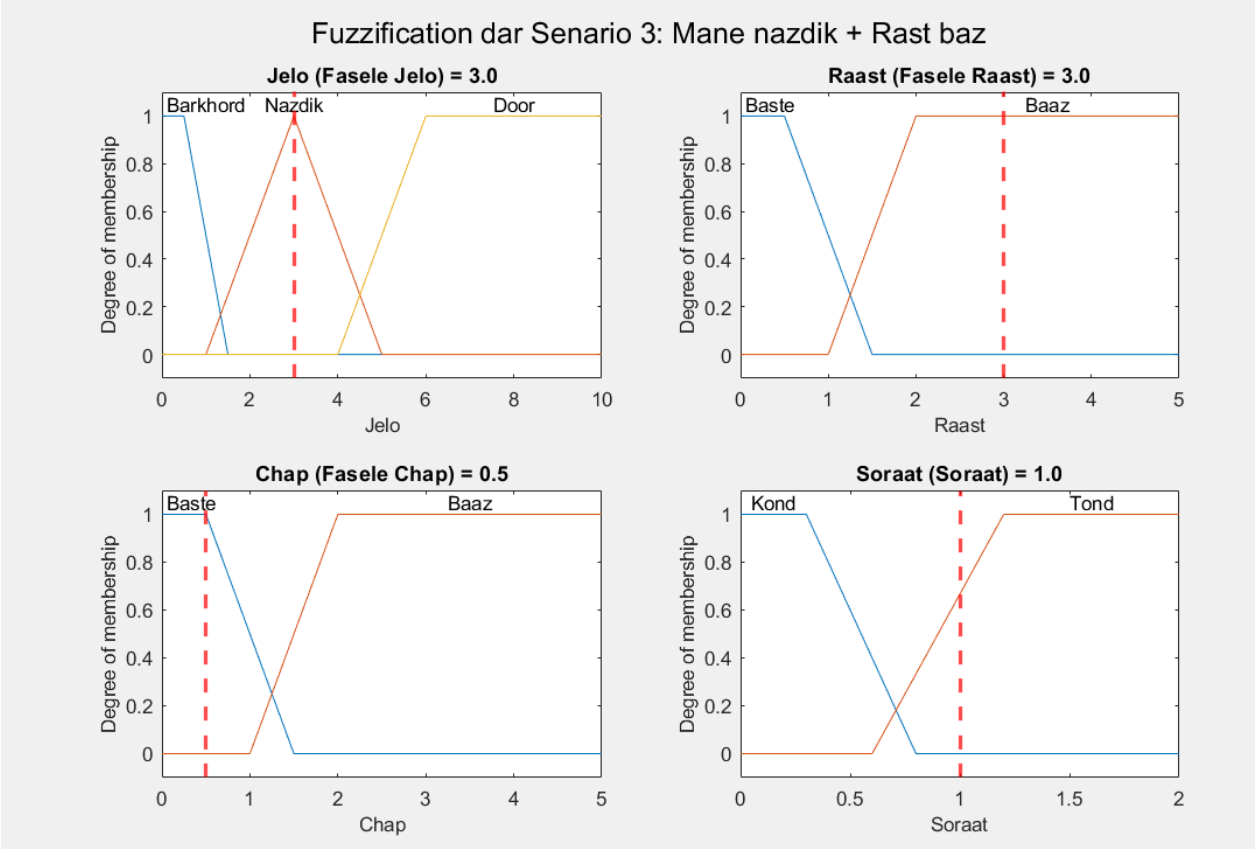

File Edit View Options
Jelo = 0.3
Raast = 3
Chap = 0.5
Soraat = 1.5
CharkheshFarmaan = 39.5
Tormoz = 95
Input: [0.3;3;0.5;1.5]
Plot points: 101
Move: left right down up
Opened system Tormoz_Ezterari, 9 rules
Help Close

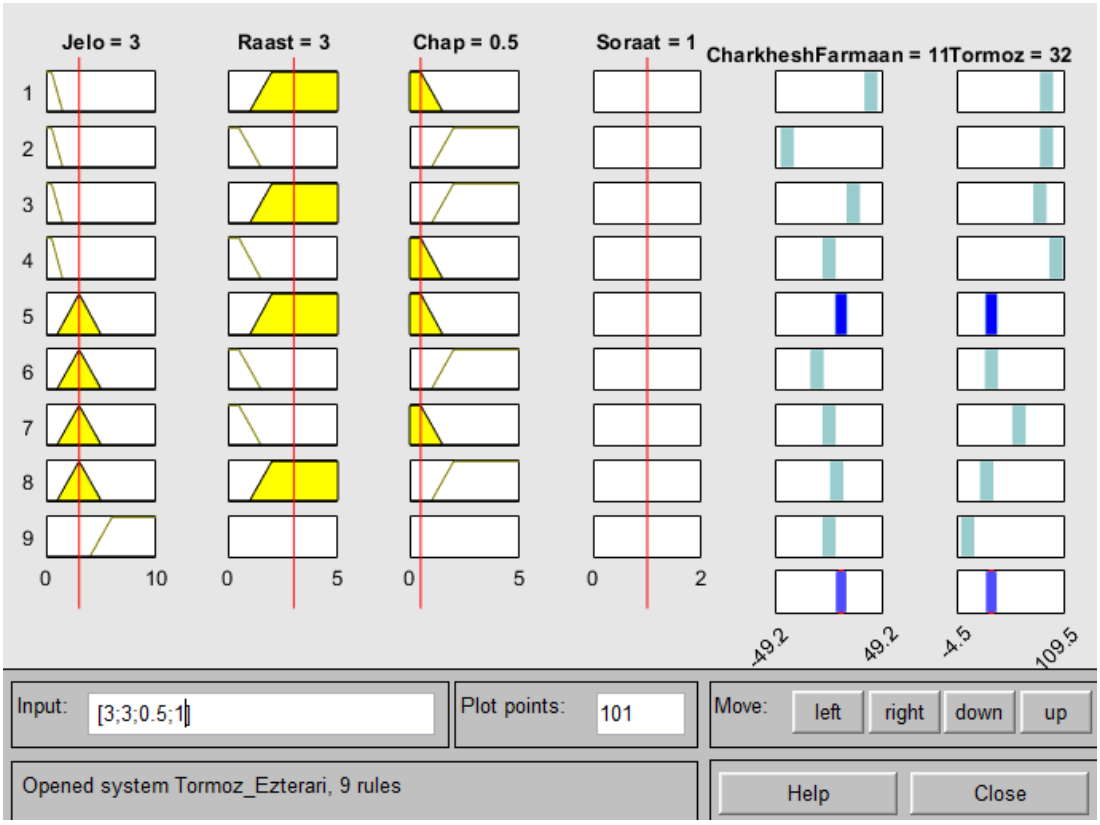
Scenario 2:


Fuzzification dar Senario 2: Barkhord + Mohasere kamel (Taleh)
Jelo (Fasele Jelo) = 0.3
Barkhord
Nazdik
Door
Degree of membership
Jelo
Raast (Fasele Raast) = 0.5
Baste
Baaz
Raast
Chap (Fasele Chap) = 0.5
Chap
Soraat (Soraat) = 0.5
Kond
Tond
Soraat

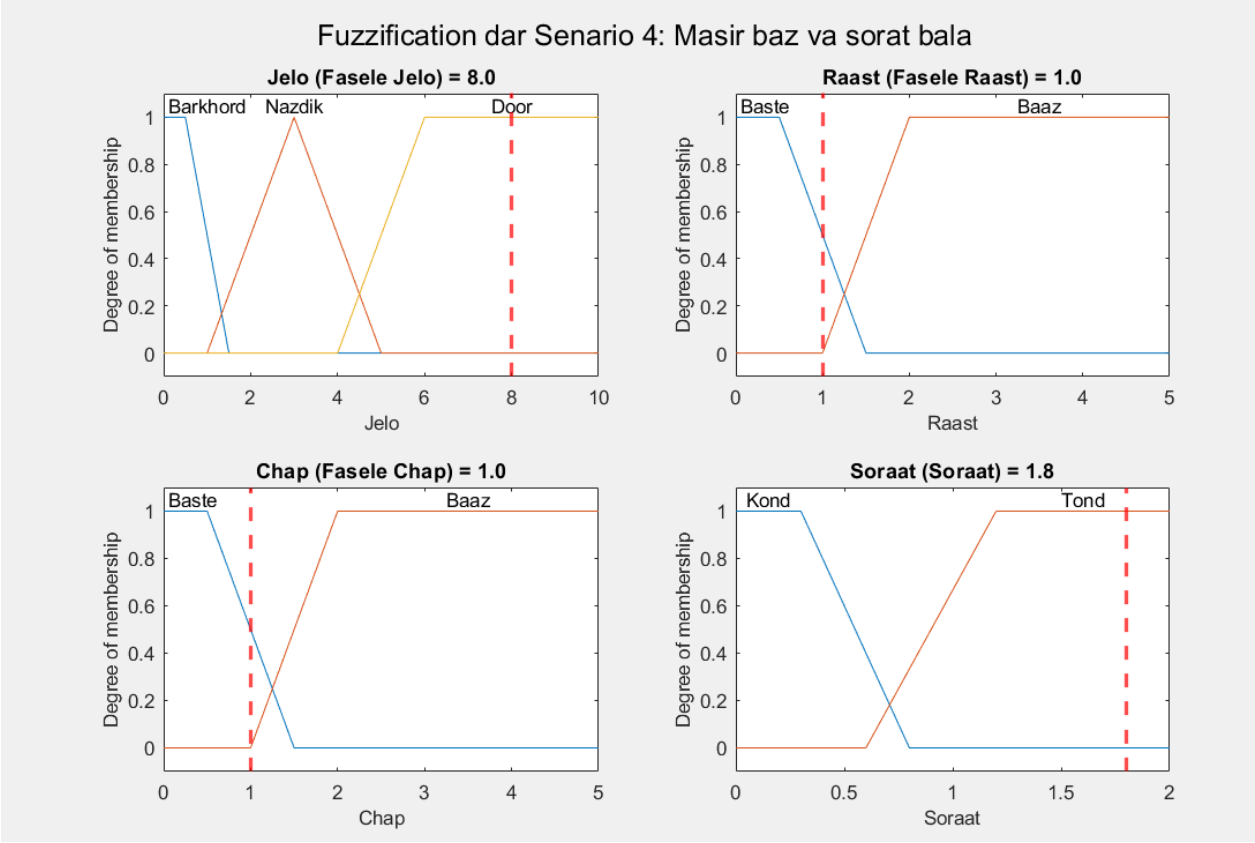

Jelo = 0.3
Raast = 0.5
Chap = 0.5
Soraat = 0.5
CharkheshFarmaan = 0
Tormoz = 100
-49.2
49.2
-4.5
109.5
Input:
[0.3;0.5;0.5;0.5]
Plot points:
101
Move:
left
right
down
up
Opened system Tormoz_Ezterari, 9 rules
Help
Close

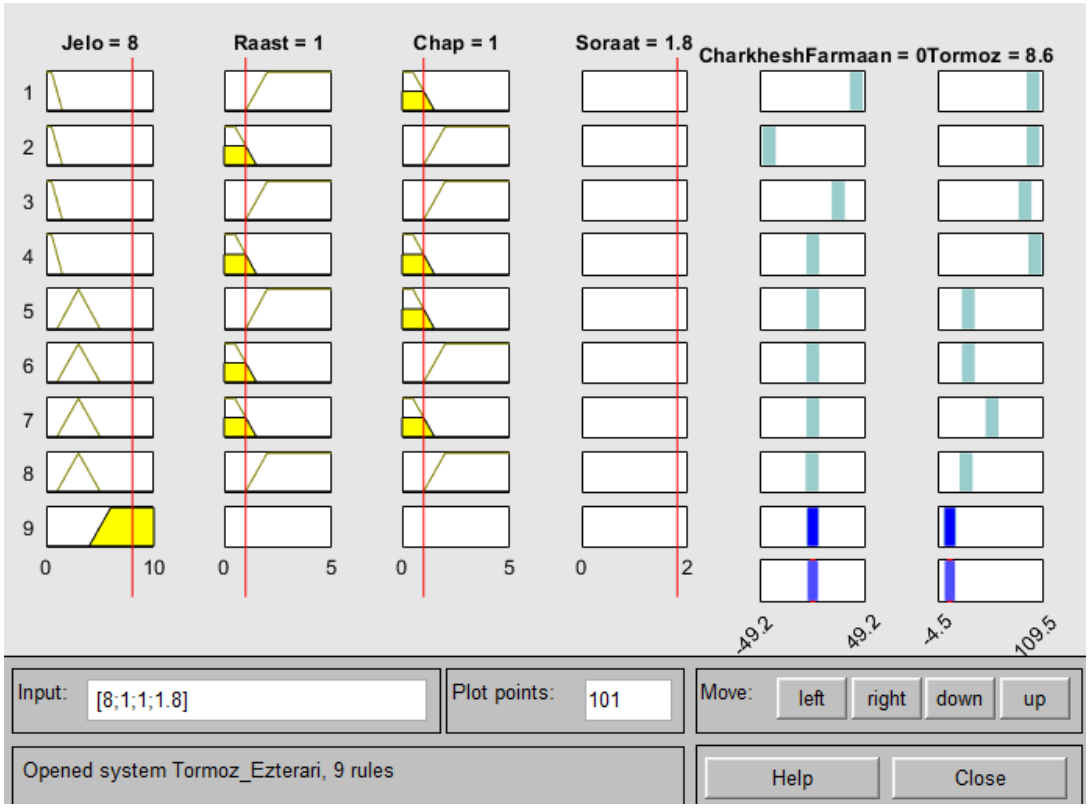
Scenario 3:

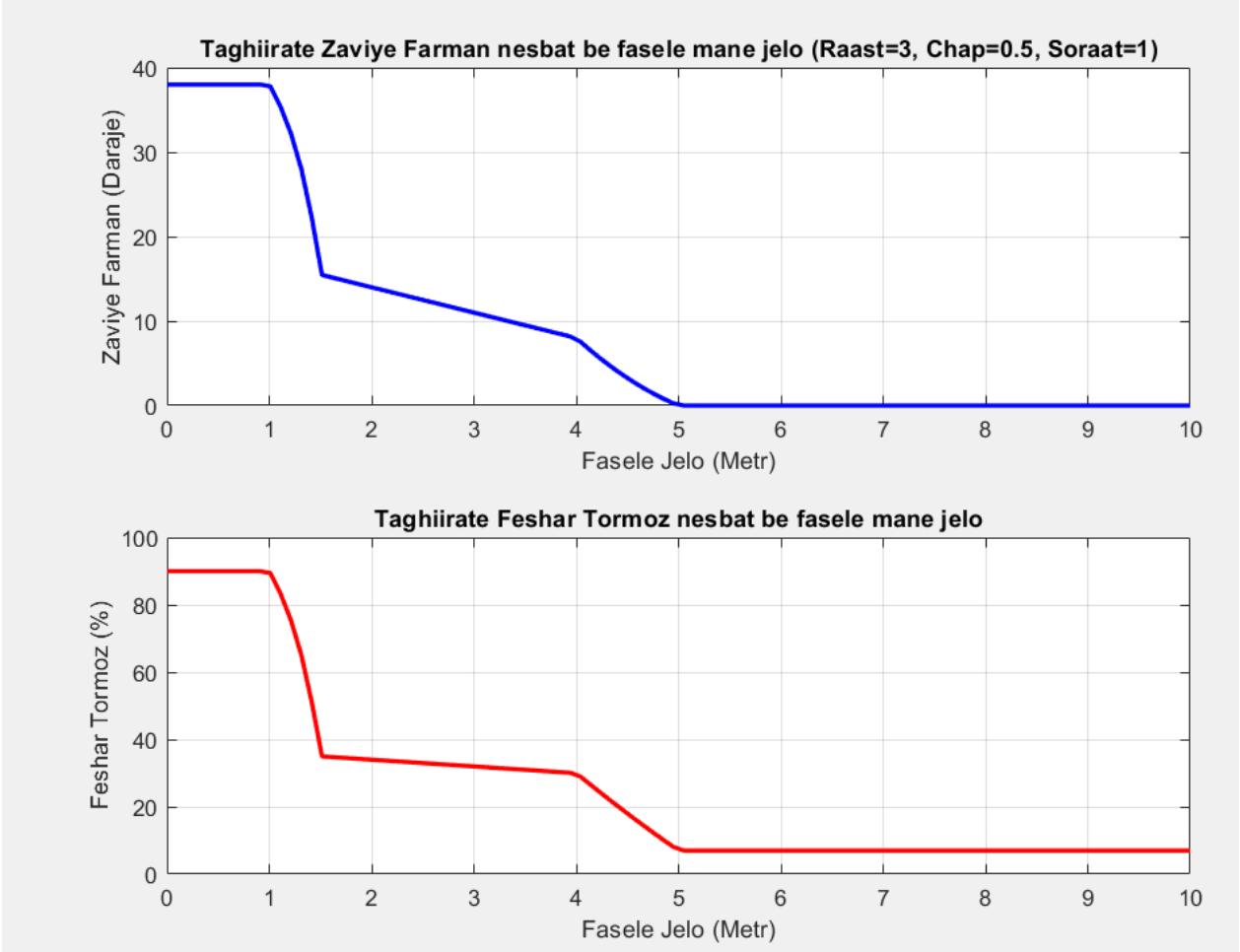

Fuzzification dar Senario 3: Mane nazdik + Rast baz
Jelo (Fasele Jelo) = 3.0
Barkhord
Nazdik
Door
Degree of membership
Jelo
Raast (Fasele Raast) = 3.0
Baste
Baaz
Raast
Chap (Fasele Chap) = 0.5
Chap
Soraat (Soraat) = 1.0
Kond
Tond
Soraat


Jelo = 3
Raast = 3
Chap = 0.5
Soraat = 1
CharkheshFarmaan = 11
Tormoz = 32
Input:
[3;3;0.5;1]
Plot points:
101
Move:
left
right
down
up
Opened system Tormoz_Ezterari, 9 rules
Help
Close

Scenario 4:


Fuzzification dar Senario 4: Masir baz va sorat bala
Jelo (Fasele Jelo) = 8.0
Barkhord
Nazdik
Door
Degree of membership
Jelo
Raast (Fasele Raast) = 1.0
Baste
Baaz
Raast
Chap (Fasele Chap) = 1.0
Chap
Soraat (Soraat) = 1.8
Kond
Tond
Soraat


Jelo = 8
Raast = 1
Chap = 1
Soraat = 1.8
CharkheshFarmaan = 0
Tormoz = 8.6
Input:
[8;1;1;1.8]
Plot points:
101
Move:
left
right
down
up
Opened system Tormoz_Ezterari, 9 rules
Help
Close


Taghiirate Zaviye Farman nesbat be fasele mane jelo (Raast=3, Chap=0.5, Soraat=1)
Zaviye Farman (Daraje)
Fasele Jelo (Metr)
Taghiirate Feshar Tormoz nesbat be fasele mane jelo
Feshar Tormoz (%)
Fasele Jelo (Metr)